High-resolution Automotive Light-field Head-Up Display with Eye-Tracking

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Abstract

A high-resolution head-up display for automotive applications has been developed. Light field display technology with eye tracking is employed for extended virtual image distance and 3D augmented perception. A field of view of 15° × 5° and an angular resolution of 76.8 pixels per degree is obtained, with a crosstalk as low as 1.7 %, yielding an broader potential for automotive applications.

Author Keywords

Head-Up Display; 3D Display; Automotive; Eye tracking.

1. Introduction

The head-up display (HUD) technology has evolved from Combiner HUD and Windshield HUD to Augmented Reality (AR) HUD [1], achieving an increasingly wider field of view (FOV) and the capability to deliver more information, thereby raising higher demands for AR-HUD capabilities. The requirement of integrating information with real-world driving scenarios cannot be adequately met by traditional HUD systems with a single fixed virtual image distance (VID). When the VID of displayed information does not align with the actual distance of the corresponding real-world scene, the human eye is forced to continuously adjust convergence between the two distances, causing driving fatigue and ultimately compromising road safety. Research has attempted to establish the relationship between VID and human visual comfort, aiming to optimize HUD display performance [2]. These efforts have indeed derived some design constraints for AR-HUD systems, contributing to more comfortable viewing experiences. However, this approach does not fundamentally resolve the core issue.

On the other hand, autostereoscopic 3D display technology has gradually advanced, evolving from multi-view displays to light-field displays and further to more promising holographic display technologies [3]. The primary goal of autostereoscopic 3D displays is to present three-dimensional effects, inherently addressing the issue of convergence. Consequently, integrated solutions combining 3D display technology with AR-HUD systems have emerged. Moreover, the problem of accommodation-convergence conflict (AC conflict) encountered in most 3D displays is reduced in HUD applications with further VID [4], making the two technologies a natural fit.

Given that the more viewpoints autostereoscopic 3D displays provide, the fewer pixels can be allocated to a single position, in the context of driving applications, since the HUD only needs to be viewed by the driver alone, the optimal choice is to add eye-tracking functionality, which enables real-time allocation of more information directly to the driver's eyes.

Lee etc. integrate autostereoscopic 3D display technology with HUD systems, proposing a comprehensive framework encompassing cylindrical lens arrays, projection optics, eye-tracking, and fusion algorithms [5]. This type of system that offers 3D information, shall it be called 3D-HUD, addresses the convergence conflict between real and virtual scenes, alleviating driver fatigue [6] while

Figure 1. Comparison of typical HUD (left) and 3D-HUD (right).

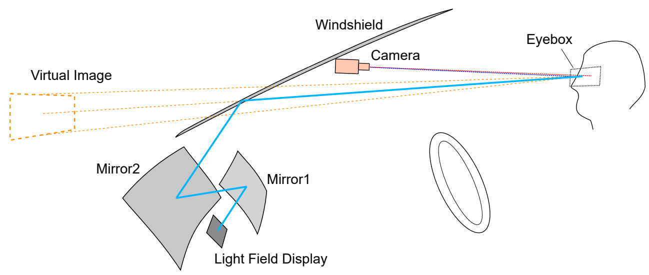

Figure 2. A general illustration of a 3D-HUD system, with a light-field display as the image source.

enhancing the immersion of virtual scenes, thereby further improving driving safety comparing to conventional AR-HUD systems, as illustrated in Figure 1.

Higher performance metrics, such as a larger field of view and higher angular resolution, are critical factors for the successful market adoption of a HUD system. In this paper, a high-resolution 3D-HUD system that utilizes light field display technology is proposed. With eye-tracking, the driver sees accurate 3D scenes. An angular resolution of 76.8 PPD allows the driver to view a clear, grain-free image, while a 15° horizontal field of view that partially covers 3 lanes ahead enables richer interactive content.

2. System Design

A LCD panel with 3840 × 2160 resolution is reflected by two free-form surface mirrors and the windshield, yielding an eye-box zone of 130 mm × 50 mm. The VID is optimized for automotive applications [2], with a field of view of 15° × 5°, yielding a virtual image with width greater than 1 m. An eye-tracking system is attached on the windshield to monitor the positions of the driver's eyes. The optical layout is shown in Figure 2.

Micro-structures are laminated on the LCD panel to form a light-field display, with a slanted angle to avoid Moiré patterns. The design parameters of the lens such as the radius, thickness, sag and pitch is carefully optimized to match the properties of the 4K panel and the viewing angle, which will be discussed later, and reduce the crosstalk at the eye-box zone.

The eye-tracking system contains a camera and a structured light

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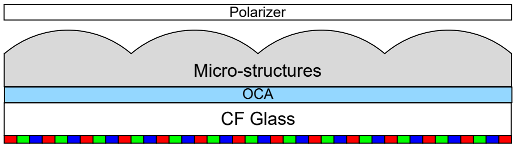


Figure 3. Illustration of stacked layers of LCD panel.

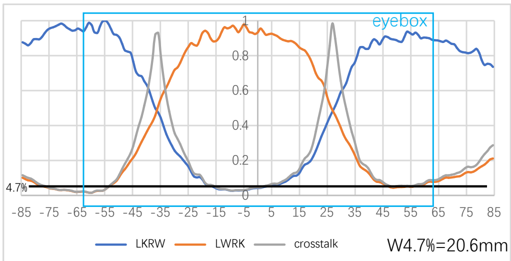


Figure 4. Crosstalk chart of the display.

source with a rapid pupil-detecting algorithm outputting positions of the driver's eyes. The 3D position of the two eyes will be sent to the light-field mapping algorithm to allocate the corresponding pixels for both eyes.

3. Experiments

Micro-structures An optimal pitch size is chosen, with a total thickness less than 500 µm. In this case, we had to remove the polarizer and thin the color filter (CF) glass of the liquid crystal (LC) box to achieve our required total optical thickness. After that, the polarizer was adhered on the top of the micro-structures to recover the function of the LCD panel. The illustration of the final structure is shown in Figure 3.

Crosstalk The result of the ray-tracing simulation shows a crosstalk in the eye-box of as low as 1.7 % and high-crosstalk slope width of 20.6 mm based on a additional crosstalk of 3 %, based on the measurement method of Lee etc. [5]. The distribution of the crosstalk is shown in Figure 4.

Resolution A comparison of the viewing experience between 2K and 4K panels under the same sub-pixel configuration is simulated. Both configurations are optimized according to the panels' properties. The zoomed areas show significant difference between the two virtual images, as shown in Figure 5.

The angular resolution is defined as

$$\mathrm{AR} = \frac{3 \times \mathrm{PR}}{p \times \mathrm{FOV}} \tag{1}$$

where AR denotes the angular resolution, PR denotes the pixel resolution of the LCD panel, p denotes the number of sub-pixels under a single structure, and FOV denotes the horizontal FOV correspondingly. Based on Equation (1), the angular resolution of the system is 76.8 PPD.

We also separate the whole display area into two parts: an AR area with angular resolution of 76.8 PPD providing 3D augmented experience over real scenes and an indicator area with angular resolution of 256 PPD providing higher resolution for detailed information. The combination of the two different areas provides flexible configuration for automotive applications. The shape of the two areas are shown in Figure 6.

Pixel Mapping A total latency, from the exposure of the eye-tracking camera to the refresh of the LCD panel, of approximate 60 ms is attained after optimization. A light-field mapping algorithm that correlates viewpoint positions, the locations of the micro-structures on the panel surface, and panel pixel coordinates is employed. The algorithm treats the two free-form lenses and the windshield as an effective lens, helping reducing the complexity of the ray-tracing relationship. This approach enables precise allocation of information to the left and right eyes.

As a result of optical aberrations and machining errors, The HUD system suffers from significant imaging distortion, both statically and dynamically. Dynamic distortion is defined in the paper as the different distortion distributions, which deviate from the designed image plane, observed at different positions within the eye-box, as shown in Figure 7. The dynamic distortion causes different distributions between the two eyes and unexpected movement of the image plane when the head moves, leading to visual fatigue and failure of augmented reality.

To reduce the dynamic distortion, a calibration algorithm that correlates distributions of images planes at multiple eye positions is employed. It helps reducing the root mean square error (RMSE) of the grid to approximately 2.2 arcmin in vertical direction, which is lower than the 5 arcmin level of the vertical disparity that is free of visual comfort proposed by Tyler etc. [7], yielding a comfort viewing experience in the calibration zone.

Demonstration The system is installed in a BMW iX, as shown in Figure. 8. A captured picture of the displayed content in the optimized eye-box position is shown in Figure 9.

4. Discussion

Resolution Generally, the total thickness of the micro-structures decreases as the pitch decreases. For the 4K panel, it is almost impossible to further decrease the pitch size as it is limited by the thickness of the top glass of the LC box. As a result, the angular resolution of the display is also limited as it is closely related to the pitch size.

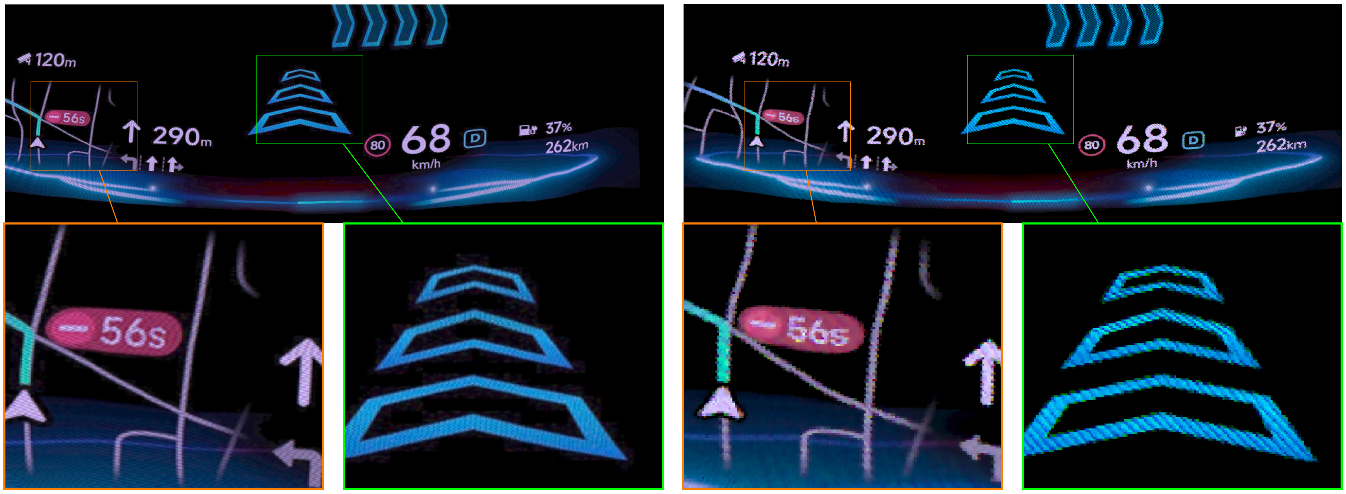
Uniformity The system has relatively low uniformity. The disuniformity is generally caused by two main factors:

1. The crosstalk caused by the aberrations of the system. It is hard to control the aberrations of the whole system with such large viewing angle. System aberrations not only affect the shape of the virtual image but also influence the distribution of crosstalk, disrupting the ideal pixel mapping relationship and resulting in localized crosstalk.
2. The rearrangement of the layers of the LCD causes decrement of the display contrast, which further amplifies the disuniformity.

5. Conclusion

A high-resolution 3D-HUD system is proposed. A 4K LCD panel with micro-structures is employed, offering 3D experience utilizing eye-tracking technology. Crosstalk as low as 1.7 % and angular resolution of 76.8 PPD are attained. Some defects such as disuniformity are to be further solved.

Through this system we explore the feasibility of applying high-resolution 3D-HUDs in actual vehicles and aim to gradually address their existing issues in subsequent stages.



(a)

(b)

Figure 5. Comparison of simulation of virtual images of the same optical layout with (a) 4K and (b) 2K panel and corresponding lens arrays.

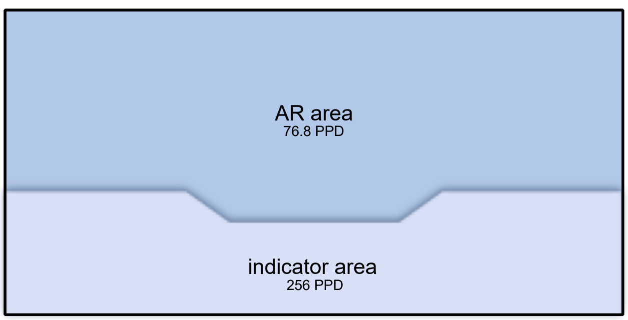


Figure 6. Illustration of two areas for different application.

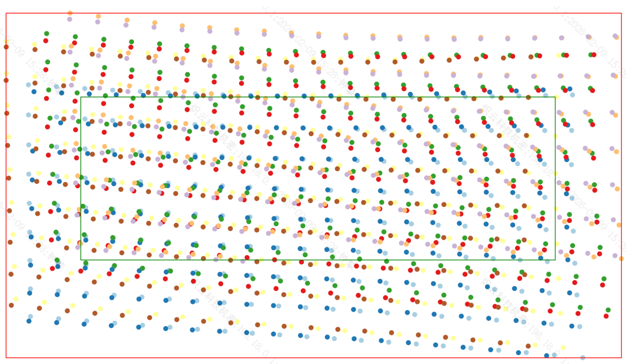


Figure 7. Illustration of the dynamic distortion. Dot grids in different colors represent distributions of a standard dot grid captured at different eye positions. The green and red rectangle shows the largest inscribed and smallest circumscribed rectangles of the visible areas.

6. References

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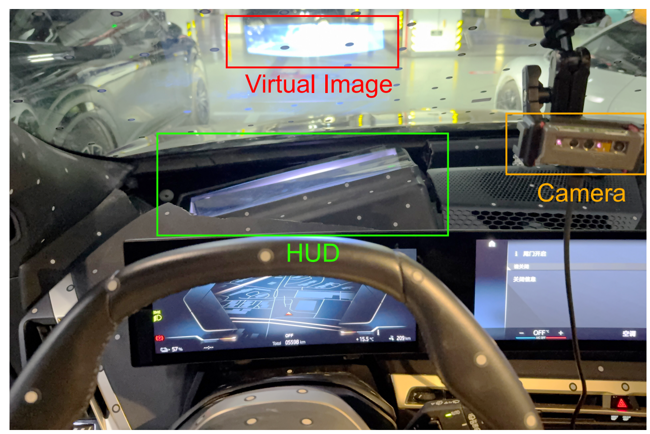

Figure 8. A view of the HUD system installed in a BMW iX.

Figure 9. A demonstration picture displayed and captured on a BMW iX. Ghost images can be seen due to the reflection of the other layer of the windshield glass.

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